STATIC and Dynamic Biplot Approaches for Socio-Economic Spatio-Temporal Analysis (Case Studies: Province of Banten, DKI Jakarta, and West Java 2020–2024)

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ABSTRACT

This study applies the *STATIS* (*Structuration des Tableaux à Trois Indices de la Statistique*) method to jointly analyse 11 socio-economic indicators across 41 regencies and cities in Banten, DKI Jakarta, and West Java over 2020–2024, capturing the structural impact of the COVID-19 pandemic and an uneven post-pandemic recovery. Demographic weights derived from population figures ensure a metrically consistent comparison space. Inter-structure evaluation via RV coefficients confirms exceptionally high longitudinal consistency (all pairwise $RV > 0.90$; $p < 0.05$), demonstrating that the regional development hierarchy remained structurally stable despite pandemic fluctuations. The STATIS compromise explains 71.88% of total variance: Dimension 1 captures socio-economic capacity (correlated with GRDP and HDI), while Dimension 2 reflects labour-market dynamics (open unemployment rate). Dynamic Biplot trajectories identify three regional patterns: V-shaped recoveries in service-based metropolitan centres, stagnant or declining trajectories in southern regencies, and progressive growth in buffer-zone cities. K-Means clustering ($k = 3$) partitions the 41 regions into *Advanced* (13), *Developing* (16), and *Underdeveloped* (12) clusters, with strong alignment to UNDP HDI normative categories. These results indicate that regional inequality is structurally persistent, necessitating targeted rather than uniform policy intervention.

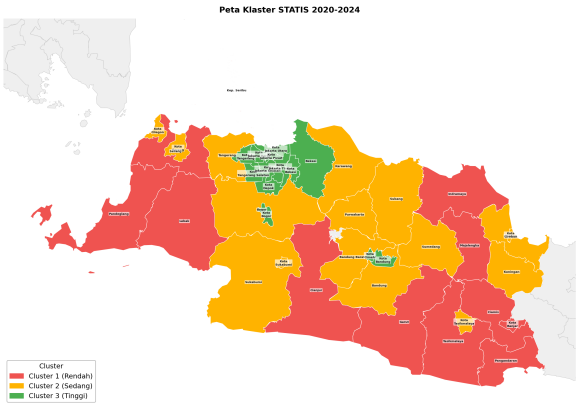


Figure 1: Spatial distribution of K-Means clusters ($k = 3$) across 41 regencies/cities ($k = 3$, 2022).

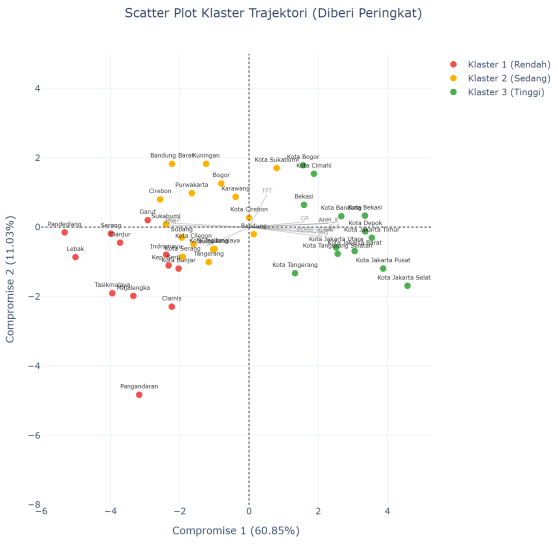


Figure 2: Compromise biplot scores with cluster overlays and trajectory arrows ($k = 3$).

CLUSTER PROFILES (STATIS Compromise, $k = 3$, 2022)

Table 1 summarises the mean indicator profiles per cluster under the STATIS Compromise (ACT-Classic, $k = 3$, 2022). GRDP and mean years of schooling increase monotonically from Cluster 1 to Cluster 3, while poverty rates follow the inverse pattern. Unemployment (TPT) is notably higher in Clusters 2 and 3 than in Cluster 1, consistent with a structural labour-market paradox in service-led metropolitan economies.

Indicator	**Cluster 1** *Underdeveloped* ($n = 12$)	**Cluster 2** *Developing* ($n = 16$)	**Cluster 3** *Advanced* ($n = 13$)
GRDP (billion IDR)	$27{,}370 \pm 18{,}004$	$58{,}076 \pm 50{,}787$	$215{,}387 \pm 158{,}352$
Poverty rate (%)	9.79 ± 2.45	8.85 ± 2.44	4.68 ± 1.39
Mean yrs schooling	7.70 ± 0.69	8.64 ± 1.02	11.14 ± 0.59
Unemployment (%)	6.55 ± 2.60	8.44 ± 1.06	8.22 ± 1.69

Table 1: Mean $\pm$ SD of key indicators per cluster (Kompromi STATIS, Scenario 1, $k = 3$, reference year 2022).

Keywords : Interactive Visualization; K-Means Clustering; Longitudinal Data; PCA; Three-Way Data Analysis

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